

Progress Report — Week 11

Course: 814601S / 814311A | **Student:** Arslan Sultan | **Student number:** 1.2.246.562.24.35253236476 | **Email:** arslan.sultan@student.oulu.fi
Employer: Universidad Politécnica de Madrid (UPM) | **Supervisor:** Juan Garbajosa
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Reporting period

10 July 2026 – 16 July 2026

Hours worked

Date	Day	Hours	Activity summary
10 Jul 2026	Fri	8	Mixed-backlog generator with feature/bug/refactor/spike mix
13 Jul 2026	Mon	8	Task dependencies and blocker propagation
14 Jul 2026	Tue	8	Defect rework spillover across sprints
15 Jul 2026	Wed	8	Sprint-phase modifiers and Scrum role modifiers
16 Jul 2026	Thu	8	pytest suite for metrics and simulation; Monte Carlo validation
Total		40	

Work completed

- Implemented mixed backlog generation with configurable task-type proportions and explicit carry-over reporting.
 - Added task dependencies with blocker propagation when prerequisite tasks are incomplete.
 - Modelled defect rework spillover affecting subsequent sprint capacity.
 - Added sprint-phase modifiers (planning, execution, review) and lightweight Scrum roles (PO, SM, Developer, Tester).
 - Built pytest test suite in `tests/` covering metrics formulas, monotonicity checks, and simulation smoke tests.
 - Ran Monte Carlo and sensitivity experiments to validate parameter changes produce consistent outcomes.
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Connection to learning objectives

- Applied software testing and quality assurance knowledge from university courses.
 - Demonstrated independent problem-solving on complex simulation logic (dependencies, rework).
 - Worked in a goal-oriented manner toward a validated, test-covered prototype.
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Challenges and how they were handled

Dependency logic introduced edge cases with circular blockers. I added validation on backlog generation to prevent impossible dependency cycles and tested with fixed seeds.

Plan for next week (17–23 Jul 2026)

- Prepare CI/HCOMP demo and poster submission materials.
 - Create improvement roadmap document (non-thesis product plan).
 - Record demo script and architecture figure for submission package.
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Student signature: Arslan Sultan — Date: 17 July 2026

Supervisor acknowledgment: Juan Garbajosa